

# AI531C — Week 10 Pre-Class Activity (PCM-10: Two Drawings, Every Network)

Prepares: Lecture 26 (Residual, encoder-decoder & transfer, Tue 13 Oct) · Lectures 27–28 (Image-to-image datasets; Autoencoders & U-Net), Wed 14 Oct · Tutorial T10 (Tue 13 Oct).

## 1 · Topic

The two diagrams that build everything ahead: the residual block ( $y = F(x) + x$ ) and the encoder-decoder with skips — plus the transfer-learning shortcut.

## 2 · Why you are preparing this

Lecture 26 closes Unit III with the two structural patterns Units IV–V are made of. Reading them the night before means class can spend its time on WHY they work and on placing YOUR project in the freeze/fine-tune quadrants.

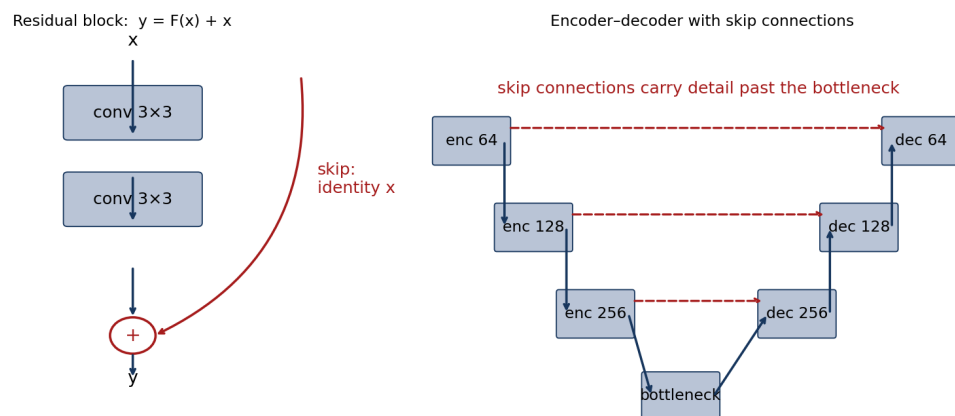
## 3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 26 chapter, the sections on residual blocks and on encoder-decoder skips (about 3 pages);
- Lecture 27 chapter, ONLY the degradation-card page (Wednesday builds datasets from it).

*Optional depth: Chollet Ch. 9 (functional-API architectures), figures only.*

## 4 · Interpret one figure (4 min)



Source: course figure, Lecture 26 (residual block; encoder-decoder with skips).

In two sentences: trace where the identity path enters the residual block's output, and state what the dashed skip lines carry ACROSS the encoder-decoder that the bottleneck cannot.

## 5 · One small calculation (5 min)

A decoder stage receives 64 channels from below and concatenates a 64-channel skip. Output channels into the next conv? If that conv is  $3 \times 3 \rightarrow 64$  with bias, count its parameters.

## 6 · Predict (4 min)

Predict: if  $F(x)$  learns to output exactly zero, what does a residual block compute — and why might 'easy to do nothing' help very deep networks?

## 7 · Inspect this fragment (3 min)

```
y = conv_block(x) + x # residual
d = upsample(deeper)
```

```
d = concat([d, skip]) # why not add?
```

Question: concat vs add for skips — what information does concatenation preserve that addition merges away?

## 8 · Bring one question to class

One question about the two architectures — or your project's dataset-size estimate (the freeze/fine-tune quadrant needs it Tuesday).

**Submit: items 4-6 — half a page maximum, before Tue 13 Oct, 9:00 AM.**

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.